

Derivatives and Rates of Change

The derivative of a function measures how quickly its output changes as its input changes ? the slope of the function at a point. For a loss function that scores how wrong a model's prediction is, the derivative with respect to a model parameter tells us whether increasing that parameter would make the loss better or worse, and by how much.

Partial Derivatives and Gradients

A model usually has many parameters at once, not just one. A partial derivative measures the rate of change with respect to a single parameter while holding all others fixed. Collecting every partial derivative into one vector gives the gradient, which points in the direction of steepest increase of the function ? and its negative points toward the steepest decrease.

The Chain Rule

The chain rule lets us compute the derivative of a function that is built by composing simpler functions, by multiplying the derivatives of each step together. A deep neural network is exactly such a composition ? layer after layer applied in sequence ? so the chain rule is what makes it possible to compute how the final loss depends on a parameter buried many layers deep, which is what backpropagation does.